

# Multiband Wire Antenna

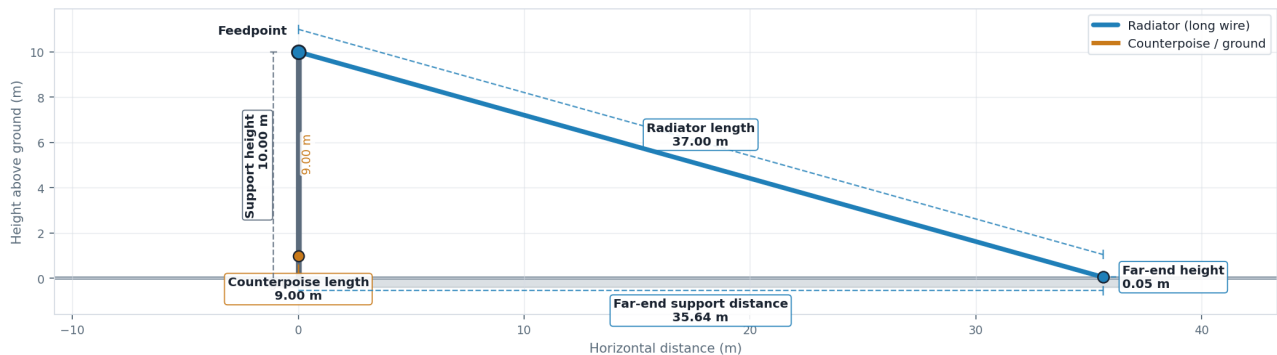
NEC2-Optimized Configuration Datasheet

37.00 m

RADIATOR LENGTH

## Configuration Overview

|                                 |                                                                    |
|---------------------------------|--------------------------------------------------------------------|
| Radiator length                 | 37.000 m                                                           |
| Counterpoise length             | 9.000 m (vertical)                                                 |
| Radiator height                 | 10.00 m                                                            |
| Recommended UnUn ratio          | 16 : 1                                                             |
| Active bands                    | 40m, 20m, 10m                                                      |
| Optimization mode               | NEC2                                                               |
| Combined score (lower = better) | 4.055                                                              |
| Wire geometry                   | Wire geometry : SLOPED (feedpoint z=10.000 m → far end z=0.0500 m) |



# Performance Report

## Per-Band Performance

| Band | Freq (MHz) | VSWR | Rating         |
|------|------------|------|----------------|
| 40m  | 7.100      | 4.21 | ★★★ EXCELLENT  |
| 20m  | 14.175     | 3.92 | RESONANCE RISK |
| 10m  | 28.500     | 2.73 | ★ MARGINAL     |

## Per-band impedance (antenna side and transmitter side):

| Band | R_ant ( $\Omega$ ) | X_ant ( $\Omega$ ) | R_tx ( $\Omega$ ) | X_tx ( $\Omega$ ) | Source |
|------|--------------------|--------------------|-------------------|-------------------|--------|
| 40m  | 291.9              | +544.5             | 32.43             | +60.50            | NEC2-V |
| 20m  | 1765.5             | -1.1               | 196.17            | -0.12             | NEC2-V |
| 10m  | 1163.3             | -254.3             | 129.26            | -28.26            | NEC2-V |

# BEST CANDIDATE — DETAILED BREAKDOWN

|                          |                                                                    |
|--------------------------|--------------------------------------------------------------------|
| Wire length              | 37.000 m                                                           |
| CP length                | 9.000 m (vertical)                                                 |
| Wire geometry            | Wire geometry : SLOPED (feedpoint z=10.000 m → far end z=0.0500 m) |
| Combined score           | 4.0553                                                             |
| VSWR penalty             | 1.4138                                                             |
| Avoidance (active bands) | 0.1073                                                             |
| Avoidance (all bands)    | 0.1073                                                             |
| NEC2 data used           | YES                                                                |

| Band | MHz    | VSWR          | Quality        | Avoidance |
|------|--------|---------------|----------------|-----------|
| 40m  | 7.100  | 4.21 MARGINAL | ★★★ EXCELLENT  | 0.2500    |
| 20m  | 14.175 | 3.92 MARGINAL | RESONANCE RISK | 0.0022    |
| 10m  | 28.500 | 2.73 GOOD     | ★ MARGINAL     | 0.0697    |

## UnUn RATIO ANALYSIS (for best antenna geometry)

|                       |                                                                |
|-----------------------|----------------------------------------------------------------|
| UnUn ratio (this run) | 16.0:1                                                         |
| Continuous optimum    | 14.52:1 (agg. VSWR penalty: 3.3621)                            |
| Best standard ratio   | 16:1 (agg. VSWR penalty: 3.3928)                               |
| Recommendation        | → Current ratio 16:1 is already optimal among standard values. |

### Standard ratio sweep:

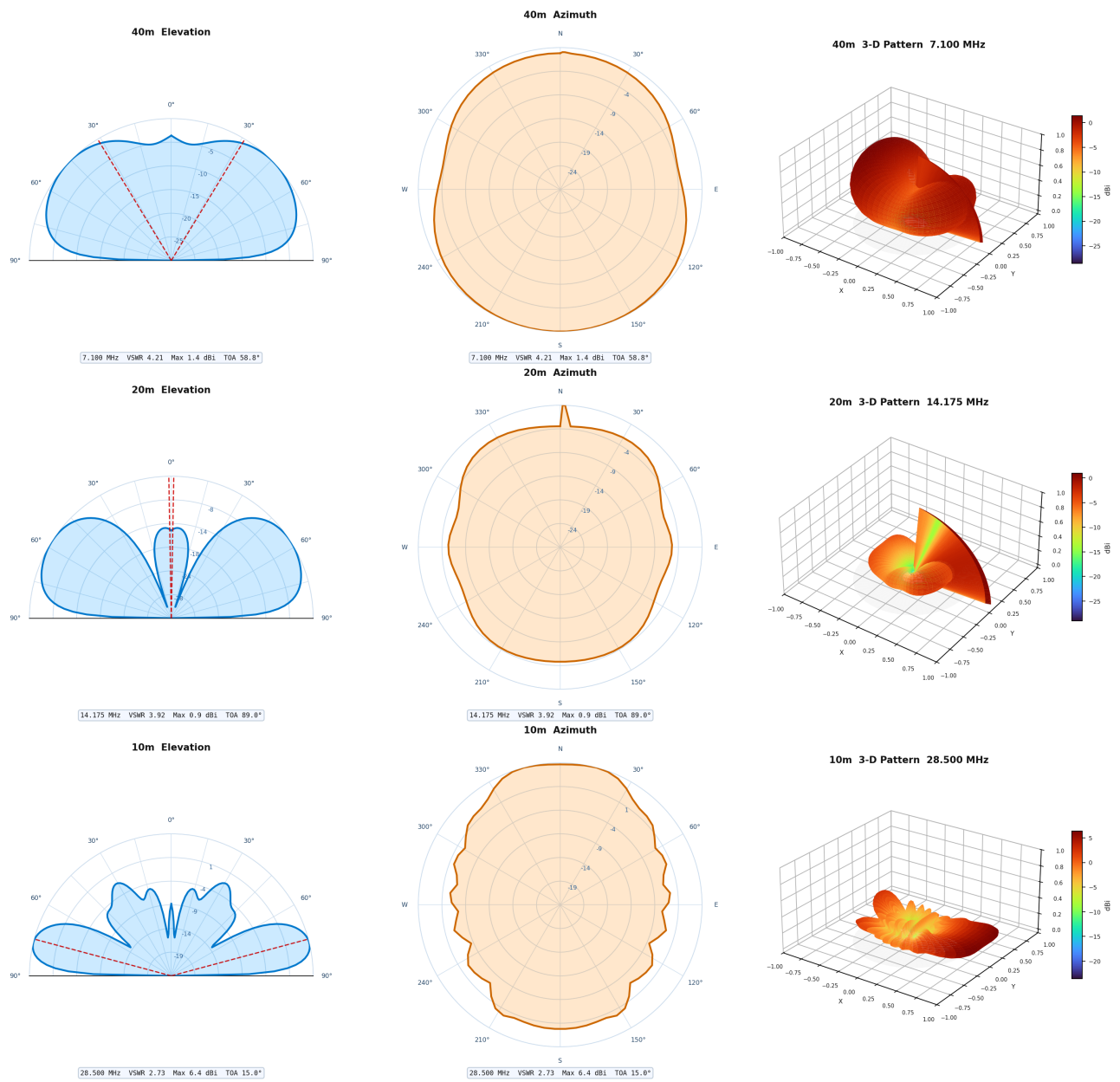
| Ratio | Score   | VSWR@40m | VSWR@20m | VSWR@10m |
|-------|---------|----------|----------|----------|
| 1:1   | 16.6401 | 26.29    | 35.31    | 24.38    |
| 1.5:1 | 14.4436 | 17.63    | 23.54    | 16.25    |
| 2:1   | 12.8883 | 13.34    | 17.66    | 12.19    |
| 3:1   | 10.7057 | 9.12     | 11.77    | 8.13     |
| 4:1   | 9.1694  | 7.08     | 8.83     | 6.10     |
| 6:1   | 6.7526  | 5.19     | 5.88     | 4.08     |
| 9:1   | 4.1237  | 4.21     | 3.92     | 2.73     |
| 12:1  | 3.4814  | 3.98     | 2.94     | 2.06     |
| 16:1  | 3.3928  | 4.13     | 2.21     | 1.58     |
| 25:1  | 4.4414  | 5.13     | 1.41     | 1.25     |
| 27:1  | 4.7775  | 5.41     | 1.31     | 1.28     |
| 36:1  | 5.9880  | 6.75     | 1.02     | 1.60     |
| 49:1  | 7.1722  | 8.81     | 1.39     | 2.13     |
| 64:1  | 8.3662  | 11.28    | 1.81     | 2.77     |

### Antenna-side impedance (independent of UnUn ratio):

| Band | R_ant ( $\Omega$ ) | X_ant ( $\Omega$ ) | Z_ant  ( $\Omega$ ) | $\theta$ (°) | Best ratio |
|------|--------------------|--------------------|---------------------|--------------|------------|
| 40m  | 291.9              | +544.5             | 617.8               | +61.8        | 12.36:1    |
| 20m  | 1765.5             | -1.1               | 1765.5              | -0.0         | 35.31:1    |
| 10m  | 1163.3             | -254.3             | 1190.8              | -12.3        | 23.82:1    |

# Radiation Patterns

Radiation Diagrams — Wire 37.00 m / CP 9.00 m (vertical)



Generated automatically by the NEC2 Antenna Length Optimizer (LU3VEA, CC0 v1.0). All values are simulation estimates over average ground; on-site results may vary.